World Action Models: The Next Frontier in Embodied AI

⚙️ Metadata

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📜 Abstract

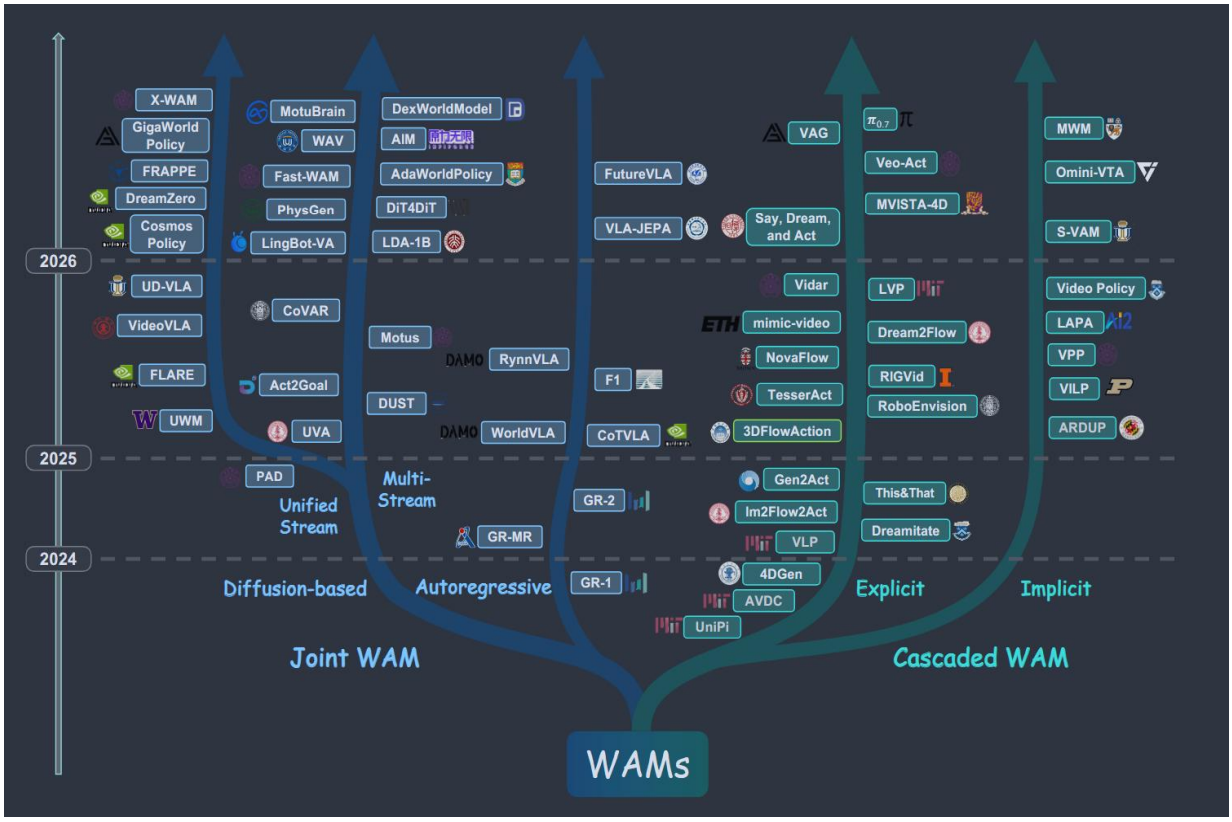
Vision-Language-Action (VLA) models have achieved strong semantic generalization for embodied policy learning, yet they learn reactive observation-to-action mappings without explicitly modeling how the physical world evolves under intervention. A growing body of work addresses this limitation by integrating world models, predictive models of environment dynamics, into the action generation pipeline. We term this emerging paradigm World Action Models (WAMs): embodied foundation models that unify predictive state modeling with action generation, targeting a joint distribution over future states and actions rather than actions alone. However, the literature remains fragmented across architectures, learning objectives, and application

scenarios, lacking a unified conceptual framework. We formally define WAMs and disambiguate them from related concepts, and trace the foundations and early integration of VLA and world model research that gave rise to this paradigm. We organize existing methods into a structured taxonomy of Cascaded and Joint WAMs, with further subdivision by generation modality, conditioning mechanism, and action decoding strategy. We systematically analyze the data ecosystem fueling WAMs development, spanning robot teleoperation, portable human demonstrations, simulation, and internet-scale egocentric video, and synthesize emerging evaluation protocols organized around visual fidelity, physical commonsense, and action plausibility. Overall, this survey provides the first systematic account of the WAMs landscape, clarifies key architectural paradigms and their trade-offs, and identifies open challenges and future opportunities for this rapidly evolving field.

Zotero links

- Local library

Notes



DEFINITION:

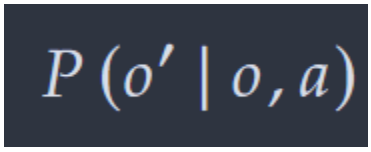
We consider an embodied agent interacting with an environment. At each timestep, the agent receives an observation o ∈ O (encompassing visual inputs, proprioceptive signals, and any other sensory modalities), a language instruction l ∈ L, and produces actions a ∈ A. We use o′ to denote the observation at the subsequent timestep.

VAM - video prediction + action generation

Video Policies - structural heritage - use generative video architectures to extract spatiotemporal representations

World Models

1.Action-conditioned World Models - environment's evolution to agent's actions



·Explicit Pixel-level Prediction - ACVP, CDNA, Deep Visual Foresight, SV2P, MCnet, ContextVP

a. Autoregressive video world models - an important paradigm of generative world models - iVideoGPT, Genie(latent action model to infer action variables from unlabeled video clips)

b. Diffusion-based video world models - explicitly modeling the distribution over possible future observations, learning the conditional distribution over future trajectories and capture multimodal environment dynamics - Diffusion World Model

·Implicit Latent-space Dynamics Model - Dynamics in a compact latent space

Recurrent State-Space Model(RSSM) - PlaNet, Plan2Explore, Dreamer

Transformer State-Space Model - TransDreamer

Predictive Representation Learning - "modeling latent embeddings of missing or future observations" (Wang 等, 2026, p. 9)

JEPA, MC-JEPA， V-JEPA 2, LeWorldModel, A-JEPA

2. Language-conditioned World Models

$$P(o' \mid o, l)$$

Early: GAN-based : MoCoGAN, TGAN, DVD-GAN

Diffusion-based : U-Net

Transformer: ViT - Sora Latte

VAEs - VideoGPT

DiT + Flow Matching - OpenAI, Kling 3, Veo 3

3. Embodied World Model

Enabling world models to acquire physics dynamics from human operation videos - Genie (latent action model), SWIM (for manipulation), DreamDojo (continuous latent actions), DexWM (Extracts 3D hand keypoints from egocentric videos)

Enhance quality of robotic demonstrations - RoboDreamer (inherent compositionality of natural language), IRASim (frame-level action conditioning module, action vector into spatial attention layer), MiLA (coarse-to-fine strategy, joint denoising correction flows + temporallly progressive denoising schedule), Ctrl-World (concatenate images from multiple cameras along token dimension within a spatial transformer), RoboScape (adaptive keypoint dynamics learning + dual-branch co-autoregressive transformer for coordinated generation of RGB and depths streams), WoW

Broaden the capabilities of world models - VT-WM, PointWorld

World Model for VLA

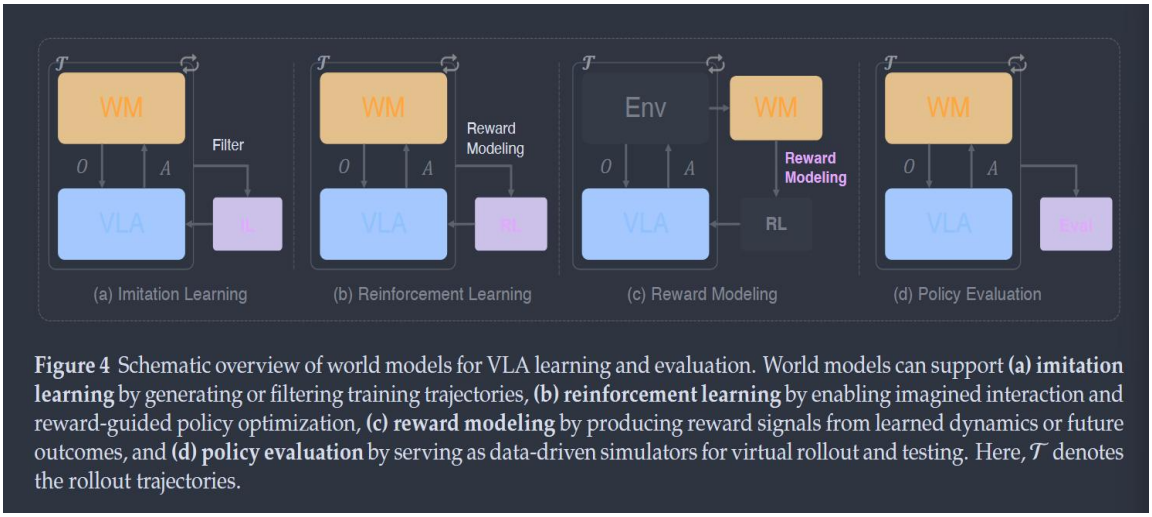


Figure 4 Schematic overview of world models for VLA learning and evaluation. World models can support **(a) imitation learning** by generating or filtering training trajectories, **(b) reinforcement learning** by enabling imagined interaction and reward-guided policy optimization, **(c) reward modeling** by producing reward signals from learned dynamics or future outcomes, and **(d) policy evaluation** by serving as data-driven simulators for virtual rollout and testing. Here, $\mathcal{T}$ denotes the rollout trajectories.

1.World model for learning

a. Imitation Learning : Generate diverse training data - DREMA (A compositional manipulation world model by combining object-centric Gaussian Splatting with a physics simulator, Ctrl-World, RoboScape

b. Reinforcement Learning : supporting safe and scalable policy learning, facilitating implicit reward estimation, improving robustness of sim-to-real transfer, policy learning

-World model as surrogate environments - Dreamer

-World model for reward modeling - VIPER (Pretrained video prediction models to evaluate the compatibility between observed trajectories and the learned dynamics of successful behaviors -> action free reward), GenReward, SRPO, RoboScape-R

2. World Model for Evaluation - Ctrl-World, Veo Robotics, Interactive World Simulator

Architecture

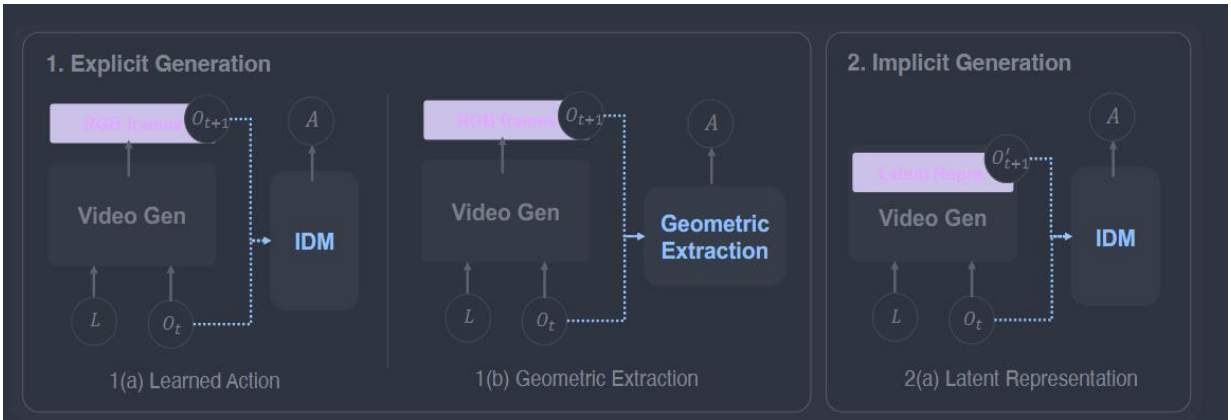


Figure 5 Schematic comparison of cascaded WAM structures. **1(a) Learned Action**: a world model generates an explicit pixel-space future plan, which is mapped to actions by a learned inverse-dynamics or action model. **1(b) Geometric Extraction**: the explicit visual plan is converted into actions or trajectories through geometric extraction. **2(a) Latent Representation**: the intermediate planning carrier is a latent future representation rather than future RGB frames, and the downstream action model decodes executable commands from it.

1. Cascaded World-Action-Model

Table 1 Comparison of Cascaded World-Action-Model (WAM) methods. **Act. Label**: whether action annotations are required. **Zero-shot**: no task-specific training/fine-tuning needed. Sim = simulation benchmark; Real = physical robot.

Method	Interm. Repr.	Stage-1 Backbone	Stage-2 Model	Act. Label	Zero-shot	Evaluation
Pixel-space Representations — Learned Action Extraction						
UniPi [6]	Pixel RGB	Video U-Net	Lightweight IDM (CNN + MLP)	✓	×	Sim: PDSketch block manip., CLIPort Real: WidowX
VLP [7]	Pixel RGB	Video U-Net + PaLM-E 12B	LAVA	✓	×	Sim: Language Table Real: Language Table robot, 7-DoF mobile arm, 14-DoF ALOHA bimanual
RoboEnvision [9]	Pixel RGB	OpenSora (DiT)	OpenSora DiT	✓	×	Sim: LanguageTable, LHMM (custom) Real: -
This&That [65]	Pixel RGB	U-Net	Custom closed-loop neural policy	✓	×	Sim: IsaacGym Real: -
Say, Dream & Act [10]	Pixel RGB	COSMOS-PREDICT2	Transformer (ACT) + Qwen2.5	✓	×	Sim: LIBERO Real: Franka 7-DoF arm
TesserAct [66]	Pixel RGB-DN	CogVideoX (DiT)	PointNet++, MLP	✓	×	Sim: RLBench Real: -
MVISTA-4D [67]	Pixel RGB-D	WAN2.2 TI2V	PointNet	✓	×	Sim: RLBench, RoboTwin2 Real: AgileX Piper
Vidar [77]	Pixel RGB	Wan2.2 (sim) / Vidu 2.0 (real)	Masked IDM (MIDM)	✓	×	Sim: RoboTwin 2.0 Real: Aloha
Gen2Act [68]	Pixel RGB	VideoPoet	Custom closed-loop neural policy	✓	×	Sim: - Real: Mobile manipulation robot
Veo-Act [78]	Pixel RGB	Veo-3	Multi-head IDM + VLA ($\pi_{0.5}$)	✓	×	Sim: IsaacLab Real: 7-DoF arm + 12-DoF dexterous hand
VAG [80]	Pixel RGB	Cosmos-Predict2	1D U-Net	✓	×	Sim: LIBERO Real: AgiBot G1, AgileX Cobot Magic
$\pi_{0.7}$ [79]	Pixel RGB	BAGEL	VLA	✓	×	Sim: - Real: Mobile manipulators, Static bimanual, UR5e
Pixel-space Representations — Geometric Extraction						
AVDC [8]	Pixel RGB → optical flow	U-Net	Off-the-shelf geometric pipeline (learning-free)	×	×	Sim: Meta-World, iTHOR Real: Franka Emika Panda
Im2Flow2Act [69]	Optical flow	AnimateDiff + Stable Diffusion (latent)	Flow-conditioned IL policy	✓	×	Sim: Custom tasks Real: UR5e
3DFlowAction [70]	3-D optical flow	AnimateDiff + Stable Diffusion	Flow-constrained optimization policy	×	×	Sim: Custom tasks Real: Franka, XTrainer
NovaFlow [71]	Optical flow	Wan 2.1 + off-the-shelf modules	Flow-based executor	×	✓	Sim: - Real: Franka Emika Panda, Boston Dynamics Spot
Dream2Flow [72]	Optical flow	Off-the-shelf I2V model	CoTrackerV3 + trajectory opt. / RL	×	✓	Sim: OmniGibson, RoboSuite Real: Franka arm, Spot, GR1 humanoid
Dreamitate [73]	Pixel RGB	U-Net	MegaPose	×	×	Sim: - Real: UFACTORY xArm 7, UR5
4DGen [74]	Pixel RGB-D + 4-D point maps	U-Net	FoundationPose + SAM2	×	×	Sim: LBM Real: Dual Franka Panda
RIGVid [75]	Pixel RGB	Kling v1.6	GPT-4o, depth estimator, FoundationPose,Grounding DINO + SAM-2,AnyGrasp	×	✓	Sim: - Real: XArm7, Aloha
LV-P [76]	Pixel RGB	WAN 2.1 I2V 14B	HaMeR, MegaSAM,Dex-Retargeting, cuRobo	×	✓	Sim: - Real: Franka Emika + parallel gripper, Unitree G1 + Inspire dexterous hand

Implicit Planning via Latent Representations						
Video Policy [13]	Latent features	U-Net	Action U-Net	✓	×	Sim: RoboCasa, Libero10 Real: Unspecified
ARDuP [82]	Latent video frames + active-region encoding	U-Net	Lightweight CNN	✓	×	Sim: CLIPort Real: WidowX
mimic-video [12]	Latent video	Cosmos-Predict2	Action Decoder DiT	✓	×	Sim: SIMPLER-Bridge, LIBERO Real: Dual Franka Emika Panda + 16-DoF dexterous hands
VPP [11]	Latent video	Stable Video Diffusion	VideoFormer + DiT Diffusion Policy	✓	×	Sim: CALVIN, MetaWorld Real: Franka Panda, Xarm + 12-DoF Xhand
VILP [81]	Latent video	U-Net + 3-D convolution	Two CNN encoders + MLP head	✓	×	Sim: Custom tasks Real: Franka Panda
LAPA [15]	Discrete latent action tokens	C-ViViT tokenizer	LWM-Chat-1M	✓	×	Sim: Language Table, SIMPLER Real: 7-DoF Franka Emika Panda
villa-X [83]	Discrete latent action tokens	Spatial-Temporal Transformer + ViT + proprio FDM	PaliGemma VLM	✓	×	Sim: SIMPLER, LIBERO Real: gripper manip., dexterous hand manip.
S-VAM [14]	Latent features	Stable Video Diffusion	Uni-Perceiver	✓	×	Sim: CALVIN, MetaWorld Real: AgileX Cobot dual-arm
OmniVTA [84]	Latent tactile features	Two-stream DiT	Diffusion Policy + MLP controller	✓	×	Sim: - Real: 7-DoF xArm + tactile sensors
MWM [85]	Latent features	DiT	Diffusion Policy	✓	×	Sim: LIBERO, RLBench Real: Franka

a. explicit planning via pixel-space representation

Learned Action Extraction - UniPi (text-conditioned spatiotemporal U-Net diffusion model)y, VLP (semantic-level intervention, equip pipeline with a VLM for hierarchical sub-action generation and tree-search-based value scoring), RoboEnvision

ThisThat, Say Dream Act (adversarial distillation to reduce denoising steps)

representation : TesserAct (depth + surface normal channels), MVISTA-4D (two-step mechanism - trajectory-level latent optimization + residual IDM refinement)

across embodiment types : Vidar, Gen2Act, Veo-Act

rather than IDM: VAG (1D U-Net as action encoder), pi0.7 (Replace second stage IDM with pretrained VLA)

Geometric Action Extraction - Replace the learned IDM with geometric computation over structured intermediate representations

AVDC (Optical Flow), Im2Flow2Act, 3DFlowAction, NovaFlow, Dream2FLow(Zero training), Dreamotate (pose tracking by MegaPose), 4DGen, RIGVid

b. Implicit Planning via Latent Representations

VPP (VAE to encode observation frames, diffusion model to single-step predict future latent sequences, a lightweight policy network conditions on these latents to produce actions), VILP (multi-view latent planning), S-VAM (directly addresses the generation quality cost)

Video Policy - ARDuP, mimic-video, MWM

“after fine-tuning SVD for video prediction, all video U-Net weights are frozen and a separate action U-Net is trained to generate action sequences conditioned on intermediate features from the video decoder, establishing feature-level conditioning as an alternative to pixel or flow inputs” (Wang 等, 2026, p. 17)2

OmniVTA - implicit latent planning for visuo-tactile domain, LAPA

2. Joint World-Action-Model

1.autoregressive generation - early hallucinations cascade into subsequent action failures

Table 2 Taxonomy-oriented summary of **Autoregressive Generation** papers reviewed in this section. “Params.” reports the main backbone scale or explicitly stated module sizes. “N/A” indicates that a clean parameter figure is not explicitly stated in the paper.

Model	Params.	Backbone	I/O Modality	Evaluation
Explicit Decoupled Representation				
GR-1 [86]	195M	GPT-style Causal Transformer	Obs, text, proprio → future patches, continuous actions	Sim: CALVIN Real: Kinova Gen2
GR-MG [87]	N/A	GPT-style + InstructPix2Pix	Obs, text, history → goal image, task progress, actions	Sim: CALVIN Real: Kinova Gen3
GR-2 [88]	30-719M	GPT-style Causal Transformer	Obs, text, proprio → future VQ tokens, continuous actions	Sim: CALVIN Real: Kinova Gen3
Unified Discrete Representations				
CoT-VLA [89]	7B	VILA-U	Obs, text → future visual tokens, discrete actions	Sim: LIBERO, Bridge-V2, Franka-Tabletop Real: -
WorldVLA [90]	7B	Chameleon-based MLLM	Obs, text → future VQ tokens, discrete actions	Sim: LIBERO Real: -
RynnVLA-002 [91]	5B	Chameleon + Action Head	Obs, text, state → future VQ tokens, continuous actions	Sim: LIBERO Real: SO100
$\mathcal{F}_1$ [93]	4.2B	Mixture-of-Transformer (MoT)	Obs, text → future VQ tokens, continuous actions	Sim: LIBERO, SimplerEnv Real: Genie-1, Franka, ARX LIFT II
Predictive Latent Representation				
VLA-JEPA [92]	2B	Qwen3-VL-2B	Obs, text → future latents, continuous actions	Sim: LIBERO, SimplerEnv, LIBERO-Plus Real: Franka

a.Explicit Decoupled Representation - GR-1, GR-MG, GR-2

b.Unified Discrete Representation - “physical dynamics and visual states are represented as unified discrete symbols, generated by the exact same next-token prediction head” (Wang 等, 2026, p. 19)

c.Predictive Latent Representations - bypass pixel-level shortcuts - VLA-JEPA - prioritize semantic abstraction and robustness

2. Diffusion-based Generation

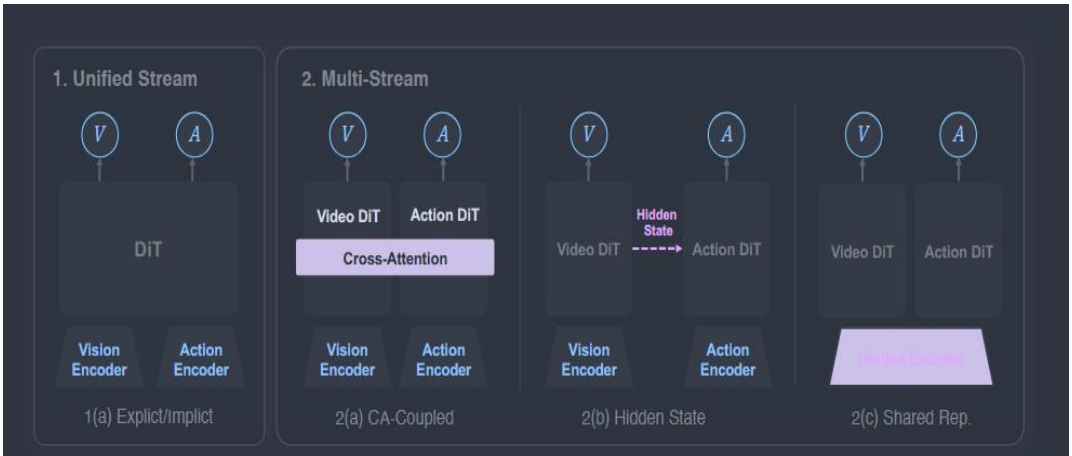


Figure 6 Taxonomy of the main architectural patterns in the diffusion-based joint WAMs. **1(a) Unified Stream**: World and action are integrated within one single DiT backbone, with world modeling realized either explicitly or implicitly. **2(a) Multi-Stream – Cross-Attention Coupled**: separate video and action DiTs are coupled through explicit cross-attention. **2(b) Multi-Stream – Hidden-State Coupling**: intermediate hidden states from the video DiT condition the action DiT. **2(c) Multi-Stream – Shared Representation**: video and action are first fused through a unified encoder before being decoded into their respective outputs.

Table 3 Taxonomy-oriented summary of **Diffusion-based Generation** papers reviewed in this section. "Params." reports the main backbone scale or explicitly stated module sizes when such information is available in the paper or the local reading reports. "N/A" indicates that a clean parameter figure is not explicitly stated in the current evidence base.

Model	Params.	Backbone	I/O Modality	Evaluation
Unified Stream: Explicit Future Prediction				
PAD [21]	N/A	ImageNet-initialized DiT	Obs, pose, optional depth, text → future image, depth, actions	Sim: MetaWorld Real: Panda
VideoVLA [94]	5B	CogVideoX-5B	Obs, text → future video latents, actions	Sim: SIMPLER Real: Realman
UWM [20]	N/A	Single DiT	Obs → future image latents, actions	Sim: LIBERO Real: Franka
DreamZero [17]	14B	Wan2.1-I2V-14B-480P	Obs, text, proprio → future video frames, actions	Sim: PolaRiS, GenieSim3.0 Real: AgiBot-G1, Franka
Cosmos Policy [16]	2B	Cosmos-Predict2-2B	Obs, text, proprio → future video latents, actions, values	Sim: LIBERO, RoboCase Real: ALOHA
GigaWorld-Policy [109]	5B	Wan 2.2-5B	Obs, text, proprio → future video latents, continuous actions	Sim: RoboTwin 2.0 Real: AgileX PiPER 6-DoF arm
X-WAM [111]	5B	Wan2.2-TI2V-5B	Obs, text, proprio → future RGB-D video, states, actions	Sim: RoboCasa, RoboTwin 2.0 Real: AC One
Unified Stream: Implicit Future Prediction				
FLARE [95]	N/A	GR00T-style policy DiT	Obs, text, proprio → future latents, actions	Sim: RoboCasa, GR-1 tabletop tasks Real: GR1 humanoid
FRAPPE [97]	1B	RDT-1B	Obs, text, proprio → future representations, actions	Sim: RoboTwin 2.0 Real: AgileX
Multi-Stream: Cross-Attention Coupled				
CoVAR [98]	N/A	Open-Sora 1.2	Obs, text, proprio → future video frames, actions	Sim: CALVIN, Libero90 Real: UR5
LDA-1B [99]	1.6B	Qwen3-VL-4B-Instruct + DINOv3-ViT-s	Obs, text → future observation, actions	Sim: RoboCasa-GR1 Real: Galbot G1, Unitree G1
DUST [101]	N/A	Eagle-2	Obs, text, proprio → future observation, actions	Sim: RoboCasa, GR-1 Real: Franka Research 3
LingBot-VA [18]	5.3B	Wan2.2-5B	Obs, text, action history → future video latents, actions	Sim: Robotwin 2.0, LIBERO Real: Franka
DexWorldModel [326]	N/A	DINOv3 + Wan2.2-5B	Obs, text, action history → future latent features, actions	Sim: RoboTwin Real: Agilex CobotMagic
AIM [102]	N/A	Wan2.2-TI2V-5B	Obs, text, action history → future RGB frames, spatial value maps, actions	Sim: RoboTwin 2.0 Real: -
Motus [19]	8B	Wan2.2-2B + Qwen3-VL-2B	Obs, text → future video latents, actions	Sim: Robotwin 2.0, LIBERO-Long Real: AC-One, Aloha-2
MotuBrain [105]	N/A	Vidu	Obs, text → future video latents, actions	Sim: RoboTwin 2.0, WorldArena Real: Humanoid platforms
AdaWorldPolicy [106]	2.8B	Cosmos-Predict2	Obs, text, proprio → future video frames, actions, forces	Sim: LIBERO-10, Variant PushT, CALVIN Real: INOVO
UD-VLA [110]	8B	Emu3	Obs, text → future visual tokens, discrete actions	Sim: CALVIN, LIBERO, SimplerEnv Real: UR5e
Multi-Stream: Hidden-State Coupled				
DiT4DiT [107]	N/A	Cosmos-Predict2.5-2B	Obs, text, proprio → future video latents, actions	Sim: LIBERO, RoboCasa-GR1 Real: Unitree G1
Fast-WAM [104]	6B	Wan2.2-5B	Obs, text → latent world features, actions	Sim: RoboTwin 2.0, LIBERO Real: Galaxea R1 Lite
WAV [100]	2.2B	Genie Envisioner	Obs, text → future visual trajectories, trajectory values, actions	Sim: LIBERO Real: Piper
Act2Goal [108]	1.76B	Genie Envisioner	Obs, goal image, proprio → future video latents, actions	Sim: Robotwin 2.0 Real: AgiBot Genie-01
Multi-Stream: Shared Representation				
UVA [96]	0.5B	Shared Transformer	Obs, optional text, action history → future image latents, actions	Sim: PushT/PushT-M, ToolHang, LIBERO-10 Real: ARX X5
PhysGen [22]	0.73B	NOVA	Obs, text, action history → future video frames, actions	Sim: LIBERO, ManiSkill Real: Franka Panda

Unified Stream - parallel generation of world action variables absorbed into a unified backbone

Explicit Future Prediction - PAD, VideoVLA, UWM, Cosmos-Predict2, DreamZero (pretrained Wan2.1 + lightweight additions, state/action encoders and an action decoder + denoising video latents and action latents), GigaWorld-Policy, X-WAM

Implicit Future Prediction - FLARE, FRAPPE

Multi-Stream

a. Cross-attention Coupling - Video DiT + Action DiT + Cross-attention - CoVAR, LDA-1B, Lingbot-VA, DexWorldModel, AIM, Being-H0.7

b. Hidden-state Coupling, DiT4DiT, Fast-WAM, WAV, Act2Goal

c. Shared Representation / Unified Encoder - UVA, PhysGen

Training data

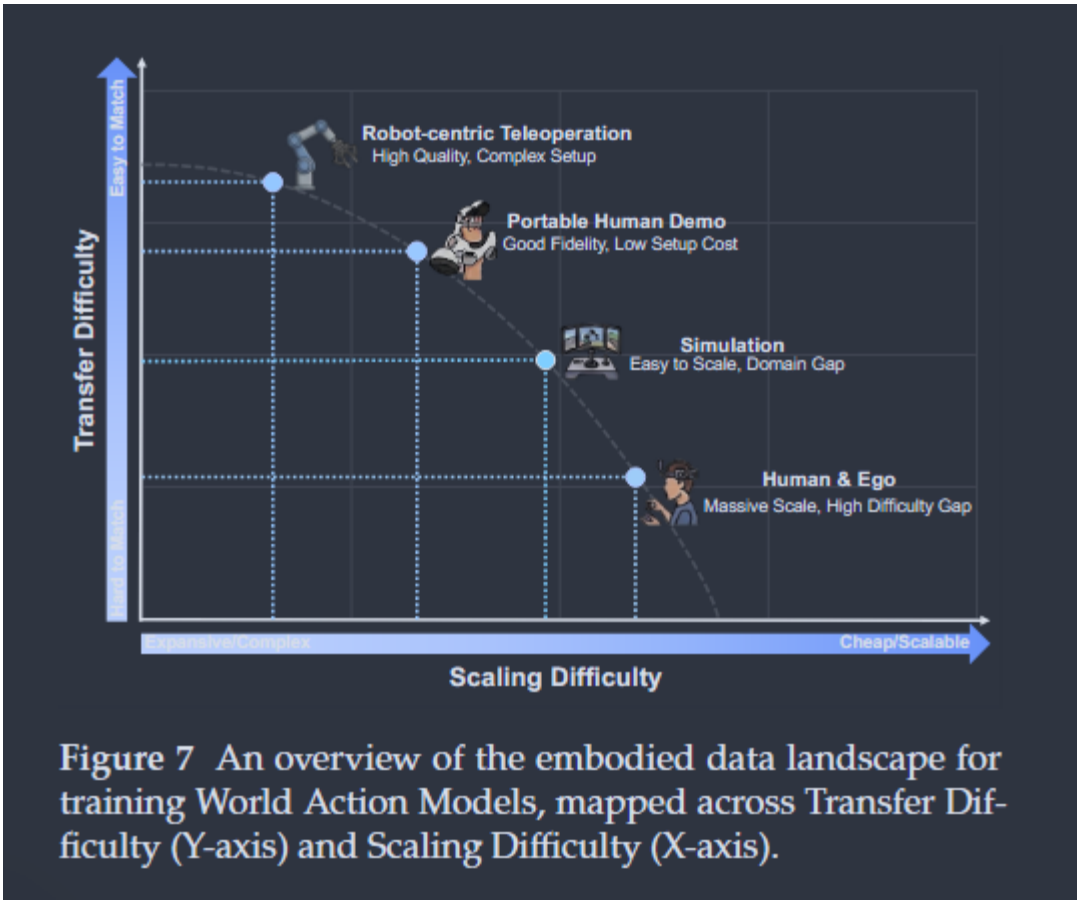


Figure 7 An overview of the embodied data landscape for training World Action Models, mapped across Transfer Difficulty (Y-axis) and Scaling Difficulty (X-axis).

1.Robot-centric Teleopration

Scaling

QT-Opt, MT-Opt, RoboNet, MIME

BridgeData, BC-Z. RT-1, Language-Table, RH20T-P

OXE, ARIO, RoboMIND, DROID, UnifoLM-WBT, Jaco Play

Multimodal and Contact-Rich Grounding

Berkeley UR5, OmniAction (Audio)

TOTO, Cable Routing Dataset, REASSEMBLE

2.Portable Human Demo (UMI style) - handheld 3d-printed gripper, wearable cameras, FastUMI

Table 5 Summary of **UMI datasets**. "Modality" denotes observation modalities: RGB, P (proprioception), D (depth), IMU, T (tactile), text.

Name	Year	Scale	Task	Env	Embodiment	Collection	Modality
FastUMI-Data [138]	2024	10K+ traj.	22	-	xArm 6	human demonstrators	RGB, P, text
FastUMI-100K [139]	2025	100K+ traj.	54	5	xArm6, Flexiv Rizon4	human demonstration	RGB, P, text
RealOmin [140]	2025	1M traj.	30	3000+	-	human demonstration	RGB, P, IMU, T, text
Hoi! [141]	2025	3,048 traj.	-	38	-	human demonstration, teleop.	RGB, D, P, T
RDT2 [142]	2026	10,000 hours	52+	100+	-	human demonstration	RGB, D, P

3.Simulation - for physics engines -

Table 6 Summary of **simulation datasets**. "Modality" denotes observation modalities: RGB, P (proprioception), D (depth), PC (point clouds), T (tactile), and text.

Name	Year	Scale	Skill	Task	Env	Embodiment	Engine	Collection	Modality
MimicGen [150]	2023	50k traj.	-	18	1	4 robots	MuJoCo, Isaac Gym	teleop., augmentation	RGB
ManiSkill2 [151]	2023	4M+ frames	4	20	-	Franka	SAPIEN	simulation	RGB, D, PC, P
RoboCasa [152]	2024	100k+ traj.	8	100	120	Franka	MuJoCo	simulation	RGB
RoboTwin [153]	2024	-	-	17	-	COBOT Magic	-	teleop.	RGB, D, P
DexMimicGen [154]	2024	21,000 traj.	-	9	-	GR1, Panda	robosuite(MuJoCo)	real2sim2real	RGB, P
QUARD-Auto [161]	2024	258,418 traj.	5	5	-	Quadruped robot	Isaac Gym	auto-collected	RGB, text
TesserAct [66]	2025	285k video clips	-	-	-	Google Robot, Franka, WidowX 250	RLBench (CoppeliaSim)	aggregation	RGB, D
RoboCerebra [155]	2025	1,000 traj.	12	100	-	-	Libero (MuJoCo)	simulation	RGB, D, text
SynGrasp-1B [156]	2025	10M traj.	2	1	-	Franka	Isaac Sim, MuJoCo	auto-collected	RGB, D, P
RoboTwin 2.0 [157]	2025	100k+ traj.	-	50	-	5 robots	Isaac Sim	simulation	RGB, D, P
TLA Dataset [158]	2025	24k tactile-action pairs	-	-	-	gripper with GelStereo 2.0	Isaac Gym + FleX	simulation	T, text
InternData-M1 [159]	2025	244k traj.	1	200	-	Franka	Isaac Gym	auto-collected	RGB, D, P
InternData-A1 [160]	2025	630k traj.	18	70	227	4 robots	Isaac Sim	simulation	RGB, P

Tactile : The TLA Dataset

Sim 2 real: synGrasp-1B, RoboTwin2.0

4. Human & Ego

Table 7 Summary of **human / egocentric datasets**. "Modality" denotes observation modalities: RGB, P (proprioception), D (depth), A (audio), PC (point clouds), T (tactile), and text.

Name	Year	Scale	Skill	Task	Env	Collection	Modality
SSv2 [162]	2018	108,499 clips	174	1	-	crowdsourced	RGB
EPIC-KITCHENS [163]	2018	55 h	-	3	-	manual	RGB, A
HowTo100M [164]	2019	136M clips	-	23K+	-	script, ASR	RGB
Kinetics-700 [165]	2019	650K clips	-	-	-	web mining	RGB
EGTEA Gaze+ [166]	2020	28.5 h	-	7	-	manual	RGB, A
Ego4D [167]	2021	3,670 h	-	5	-	manual	RGB, A
HOI4D [168]	2022	2.4M frames	-	54	-	manual	RGB, D
EgoVid-5M [169]	2024	5M clips	-	-	-	derived	RGB, text
COM Kitchens [170]	2024	145 clips	131	-	-	manual	RGB
Egocentric-10K [171]	2025	10K h	-	-	-	manual	RGB
DreamDojo-HV [35]	2026	43,827 h	6,015	6,015	-	crowdsourced	RGB, text
Assembly101 [172]	2021	4,321 videos	-	4	-	manual	RGB, P
H2O [173]	2021	571,645 frames	-	36	3	manual	RGB, D, P, PC, text
EgoPAT3D [174]	2022	1M frames	-	-	15	manual	RGB, D, PC, P
Ego-Exo4D [175]	2023	1,286 h	8	4	123	manual	RGB, A, PC, P
ARCTIC [176]	2023	2.1M frames	-	2	-	manual	RGB, D, P
HoloAssist [177]	2023	166 h	-	20	-	manual	RGB, D, P, A, text
HOT3D [178]	2024	3.7M+ frames	-	-	4	manual	RGB, PC, P
TACO [179]	2024	5.2M frames	15	3	-	manual	RGB, D, text, P
Aria Everyday Activities [191]	2024	7.3 h	-	-	5	manual	RGB, A, P, PC, IMU
OAKINK2 [181]	2024	4.01M frames	60	150	4	manual	RGB, P, text
Nymeria [182]	2024	300 h	-	20	50	manual	RGB, PC, P, A
EgoMimic [183]	2024	16 h	-	-	-	manual, teleop.	RGB, P
PH^2D [184]	2025	3M+ frames	-	4	-	manual	RGB, P
Human2Robot [363]	2025	2,600 episodes	-	14	-	teleop.	RGB, P
Humanoid Everyday [185]	2025	10.3K traj., 3M frames	221	260	-	teleop.	RGB, D, T, P, text
IndEgo [186]	2025	294 h	5	25	-	manual	RGB, P, A, PC
PLAICraft [187]	2025	10K+ h	10	8	1	crowdsourced	RGB, A
HD-EPIC [188]	2025	41.3 h	-	-	9	manual	RGB, A, P
UniHand [189]	2025	1,155 h	-	150+	-	manual, annotation	RGB, P, text
OpenEgo [364]	2025	1,107 h	-	290	600+	aggregation, annotation	RGB, D, P, text
Ego-Centric Human Manipulation [190]	2025	500,000 image-action pairs	-	12	25	manual	RGB, P, text
Kaiwu [180]	2025	11,664 actions	-	-	-	manual	RGB, D, T, A, P
EgoDex [192]	2026	829 h	-	194	-	manual	RGB, P, text
EgoScale [362]	2026	20,854 h	4	6,015	9,869	manual	RGB, P, text
EgoVerse [365]	2026	1,362 h	6	1,965	240	manual	RGB, D, P, text
HumanNet [366]	2026	1M h	8	720K+	-	aggregation, crawling, annotation	RGB, P, text

clouds and eye gaze, **OAKINK2** [181] links hierarchical task intent with fine poses, and **Nymeria** [182] provides 300 hours of full-body motion capture.

Passive World Modeling: Action Semantics and Visual Dynamics - Short horizon interactions

Bridging the Action Gap

Towards Generalist Pretraining Mixtures

Evaluation - fidelity of future states and effectiveness of generated actions

1.World Modeling Capability

Table 8 Summary of world modeling evaluation metrics and benchmarks.

Name	Year	Evaluation Focus	Metric / Implementation
Visual Fidelity			
PSNR	–	Pixel-level reconstruction fidelity	Log ratio between maximum signal value and MSE
SSIM [193]	2004	Structural similarity in luminance, contrast, and structure	Structural similarity index
LPIPS [194]	2018	Deep perceptual similarity	Weighted distance in deep feature space
DreamSim [195]	2023	Human-aligned perceptual similarity	Similarity in fused embedding space trained on human triplet judgments
DINO [196]	2023	Semantic / instance-level alignment	Cosine similarity in DINOv2 feature space
FVD [197]	2018	Distribution-level realism and temporal quality	Fréchet distance in pretrained video feature space
Object Dynamics			
VideoPhy [198]	2024	Physical interaction scenarios including solid-solid, solid-fluid, and fluid-fluid interactions; object continuity and physical plausibility	Binary human annotations on semantic adherence and physical commonsense
PhyGenBench [199]	2024	Physical commonsense alignment in generated videos	PhyGenEval with key physical phenomena detection, physics order verification, and overall naturalness
VBench-2.0 [200]	2025	Physics and commonsense violations, including mechanical, thermal, and material state changes, and abnormal entity behavior	Video-based multi-question answering for Physics; clip-level abnormal-entity detector for Commonsense
WorldModelBench [201]	2025	Physics adherence in long-horizon generated world dynamics	Five binary physical-law checks with human annotations and a fine-tuned VLM-based automatic judger
Physics-IQ [202]	2026	Future evolution of real-world physical events from conditioning frames	Spatial IoU, Spatiotemporal IoU, Weighted Spatial IoU, MSE
Motion and Trajectory Plausibility			
WorldScore [203]	2025	Controllability, quality, and dynamics; dynamics includes motion accuracy, motion magnitude, and motion smoothness	Optical-flow-based motion magnitude; smoothness via comparison to interpolated references
EWMBench [204]	2025	Motion correctness and trajectory consistency in embodied world models	EEF trajectory metrics: HSD, nDTW, DYN
Action Plausibility			
WorldSimBench [205]	2024	Whether generated situation-aware videos preserve control-relevant information for downstream manipulation	Implicit Manipulative Evaluation
Wow, wo, val! [206]	2026	Whether generated videos can be translated back into executable actions	IDM Turing Test with downstream real-world execution success

a. Visual Fidelity - PSNR, SSIM

feature-based - LPIPS, DreamSim, DINO, FVD

b. Physical Commonsense

Object Dynamics

Motion and Trajectory Plausibility - World Score, EWMBench

c. Action Plausibility - WorldSimBench, Wow, wo, val

2. Action Policy

Table 9 Summary of evaluation benchmarks for action policy. **Obs.** denotes observation modalities: RGB, Pose, D (depth), S (segmentation), PC (point clouds), T (tactile), N (Normal Vector). **Eval. Focus tags:** Generalization, Multi-task, Long-horizon, Language, Data Scaling, Sim-to-Real, Dexterity, Memory, Lifelong.

Name	Year	Obj.	Tasks	traj.	Obs.	Robots	Simulator	Eval. Focus
General Manipulation								
MetaWorld [207]	2019	80	50	-	Pose	Sawyer	MuJoCo	MT
RLBench [208]	2020	28	100	-	RGB, D, S	Franka Panda	CoppeliaSim	G MT
Robomimic [209]	2021	15	8	6K	RGB, D	Franka Panda	MuJoCo	–
Franka Kitchen [210]	2020	10	7	513	Pose	Franka Panda	MuJoCo	LH
CALVIN [213]	2021	28	34	20K+	RGB, D	Franka Panda	PyBullet	LH
ManiSkill [211]	2021	100+	4	30K+	RGB, D, S	Franka Panda	SAPIEN	G
ManiSkill2 [151]	2023	2144	20	30K+	RGB, D, S	Franka Panda	SAPIEN	MT
ManiSkill3 [212]	2025	10K+	62	-	RGB, D, S	Franka Panda	SAPIEN	DS MT
VIMA-Bench [214]	2023	100+	17	600K+	RGB, D, S	UE5	PyBullet/Ravens	Lang
VLMbench [215]	2024	22	8	6K+	RGB, D, S	Franka Panda	CoppeliaSim/RLBench	Lang
ARNOLD [230]	2023	40	8	10K+	RGB, D, S	Franka Panda	IsaacSim	G
LIBERO [216]	2023	67	130	6.5K	RGB, D	Franka Panda	MuJoCo/robosuite	LH
Libero-plus [4]	2025	-	10030	20K+	RGB, D	Franka Panda	MuJoCo	G
Libero-pro [217]	2025	-	-	-	RGB, D	Franka Panda	MuJoCo	G
Libero-x [218]	2026	60+	600	2520	RGB, D	Franka Panda	MuJoCo	G
COLOSSEUM [219]	2024	-	20	2K	RGB, D	Franka Panda	CoppeliaSim/RLBench	G
GemBench [231]	2025	20+	60	-	RGB, D	Franka Panda	CoppeliaSim	G
AGNOSTOS [220]	2025	-	41	3.6K	RGB, D, S	Franka Panda	CoppeliaSim/RLBench	G
SimplerEnv [229]	2024	-	-	-	RGB, D	Google Robot, WidowX	SAPIEN	S2R
RoboCasa [152]	2024	2509	100	100K+	RGB, D	4+	MuJoCo/robosuite	LH
GenManip [225]	2025	10	200	-	RGB, D, S	Franka Panda	IsaacSim	G
VLABench [226]	2024	2000+	100	5K	RGB, D	Franka Panda	MuJoCo	Lang
RoboVerse [222]	2025	5.5K	276	500K	RGB, D	5	MetaSim	S2R DS MT
RoboSuite [227]	2020	20	9	12K	RGB, D	10	MuJoCo	MT
RoboEval [221]	2025	-	8	3K+	RGB, D	Franka Panda	–	G Dex
RoboMME [224]	2026	-	16	1600	RGB, D, S	Franka Panda	ManiSkill3	Mem
RoboLab [228]	2026	-	120	-	RGB, D	Franka Panda	IsaacSim	–
PolaRiS [223]	2025	-	-	~350	RGB, D	Franka Panda	IsaacSim	S2R
Arms and Humanoid Form								
RoboTwin [153]	2025	10+	-	320	RGB, D, S	Aloha-AgileX	SAPIEN/ManiSkill3	MT Dex
BiGym [232]	2024	10+	40	2K	RGB, D	Unitree H1	MuJoCo	MT Dex
HumanoidBench [233]	2024	15	27	45K+	RGB, D, T	Unitree H1 + Shadow-Hand	MuJoCo	MT Dex
HumanoidGen [234]	2025	-	20	200K+	RGB, D	Unitree	SAPIEN	DS Dex
Mobile Manipulation								
HomeRobot [236]	2023	7892	12	-	RGB, D	Hello Robot Stretch	AI Habitat	G S2R
ManipulaTHOR [235]	2021	2.6K+	28	12.8K+	RGB, D, N	Kinova Gen3 on Mobile Base	Unity/AI2-THOR	G
BEHAVIOR-1K [237]	2023	5215	1K	230K+	RGB, D	Mobile Manipulator	OmniGibson	G LH
Contact and Deformation Manipulation								
SoftGym [238]	2021	4	10	10K+	RGB, D, P	Sawyer, Franka	NVIDIA FleX	Dex
PlasticineLab [239]	2021	1	10	1.2K	RGB, D, P	Rigid End-Effector	Taichi+DiffTaichi	Dex
DaXBench [240]	2023	4	9	9K+	RGB, D, P	–	DaX	Dex
TacSL [241]	2025	-	3	-	RGB, D, T	Franka Panda	Isaac Gym	S2R Dex
ManiFeel [242]	2025	-	6	-	RGB, D, T	Franka Panda	Isaac Sim/TacSL	S2R Dex
Real Device								
RoboArena [243]	2025	-	-	-	RGB, D	Franka Panda	–	G
RoboChallenge [244]	2026	-	30	-	RGB, D	4	–	MT
Maniparena [245]	2026	-	20	10812	RGB, D	X2Robot, Quanta X1	–	G LH

a.General Manipulation - RLBench, MetaWorld, Robomimic, LIBERO, ManiSkill series, RoboCasa, COLOSSEUM, LIBEROplus, GemBench

b.Bimanual and Humanoid Manipulation - RoboTwin, Humanoid Bench

c. Mobile Manipulation - ManipulaTHOR

d.Contact-rich and Deformable Object Manipulation

Tactile - TacSL, ManiFeel

e.Real-robot Evaluation - RoboArena. RoboChallenge, Maniparena

🧐 Personal Summary

💭 Thought Inspiration

> Tips: What aspects did you question, how do you think it can be improved?

📜 Conclusion

> Tips: What was done, what problem was solved, innovations and shortcomings?